

# LIANG YU

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German and South African dual citizen



## EDUCATION

### Doctor of Philosophy | *Physics*

Massachusetts Institute of Technology (MIT)

Jun 2019

Cambridge, MA, USA

- Thesis: Elucidating the Evolution Histories of Giant Planets with *K2* and *TESS*
- Research areas: Exoplanet detection and characterization
- GPA: 5.00 / 5.00

### Bachelor of Science | *Astronomy & Physics*

Yale University

May 2014

New Haven, CT, USA

- Graduated summa cum laude, with exceptional distinction in the Astronomy & Physics major
- GPA: 4.00 / 4.00

## RELEVANT EXPERIENCE

### Lead AI Architect

BMW Group

Nov 2024 – present

Munich, Germany

- Lead the development of onboard machine learning platform to support custom inference frameworks, model update, agentic workflows and monitoring on constrained in-vehicle hardware.
- Contribute to production-grade model compilation and optimization pipeline for small language models and automatic speech recognition on automotive SoCs. Design code architecture and establish CI/CD pipelines and automated testing.
- Design and conduct PoCs to evaluate emerging generative AI models, agentic frameworks and accelerators to guide future supplier nominations.
- Lead hiring and organizational development for the ML Core Platform team in India, including technical mentorship and performance management.
- Advise executive leadership and board stakeholders on AI strategy, influencing investment decisions and product direction across global markets, with deep focus on China (technology trends, competitors, and regulatory landscape).

### Data Science Team Lead, Big Loop and Advanced Systems

Volkswagen Group CARIAD SE (Audi AG until Apr 2021)

Oct 2020 – Oct 2024

Ingolstadt, Germany

- Hired and supervised a group of 15+ software engineers and data scientists. Established engineering best practices, development workflows, and a production-grade codebase for data-centric AI systems for autonomous driving.
- Led the development of in-vehicle data selection triggers for camera images, enabling efficient acquisition of high-value training data by detecting high-uncertainty and novel objects in real-world driving scenarios.
- Defined product vision and roadmap for data selection triggers, communicated with cross-functional stakeholders and oversaw successful deployment of triggers across the VW Group fleet.
- Contributed as data scientist and developer to the research, testing and optimization of uncertainty quantification and anomaly detection algorithms to solve the “long-tail” problem, as well as active learning experiments for perception tasks (semantic segmentation, object detection).
- Influenced executive decision-making by developing statistical models to estimate the required empirical test effort for autonomous vehicles.

### Data Scientist, Upstream Integrated Solutions

ExxonMobil

Aug 2019 – Sep 2020

Spring, TX, USA

- Developed smoothing filter-based models for the calibration and interpretation of time series data from acoustic sand monitors and seismic sensors, and a web API for visualizing sensor data and sending real-time alerts to users.
- Automated predictions of drilling outcomes of basins using supervised learning. Matched the accuracy of predictions by experts in geophysics.
- Researched and applied polarization filtering and spectral analysis to detect low signal-to-noise microseismic events in time series data from three-component seismic sensors.

**Graduate Research Student, Exoplanet Group**

MIT, Department of Physics

Sep 2014 – Jun 2019

Cambridge, MA, USA

- Co-authored [Astronet-Triage](#) and [Astronet-Vetting](#), convolutional neural networks that identify planet candidates and false positives in time series data from large-scale surveys. Currently used by the Transiting Exoplanets Survey Satellite (TESS) mission.
- Constructed pipeline to extract stellar flux, remove systematic noise with Gaussian process regression and identify planet-like signals from raw data taken by the K2 space mission. Discovered 81 confirmed planets and candidates.

**Intern, Big Data Strategies**

Totsa Total Oil Trading SA

Jun 2016 – Aug 2016

Geneva, Switzerland

- Researched supervised learning techniques to improve ship tracking accuracy, predict ship destinations based on partial trajectories, and identify patterns and anomalies in global ship movement.
- Developed algorithms to extract the most common ship trajectories between ports and calculate the length and duration of a typical trip.
- Simplified ship trajectories and removed outliers with clustering methods for easier processing.

**STUDENT SUPERVISION****ElHayani, M. Master's thesis — Technical University of Munich**

2023-2024

Title: Real-Time Object Detection Uncertainty Quantification Using Augmented Images for Autonomous Vehicles

**TECHNICAL SKILLS**

**AI & Machine Learning:** AI/ML Architecture, Edge AI & On-Device ML, Computer Vision, Generative AI (LLMs/VLMs/SLMs), Agentic AI, RAG, MLOps, Model Evaluation & Benchmarking, Active Learning

**AI Systems & Optimization:** Model Optimization, Quantization, NPU/GPU Acceleration, AI Workload Orchestration, ONNX/ONNX Runtime

**Data Science:** Statistical Modeling, Uncertainty Quantification, Time-Series Analysis, Predictive Modeling, Experimental Design

**Programming & Frameworks:** Python (Pandas, PyTorch, TensorFlow, Keras, Scikit-learn, LangChain, LangGraph), SQL, Bash

**Cloud & Tools:** Microsoft Azure, Git

**Languages:** fluent English and Mandarin Chinese; intermediate German, French, Spanish, and Italian

**INVITED TALKS**

**Finding the Unknown Unknowns: Intelligent Data Collection for Autonomous Driving Development**

2022

WeAreDevelopers World Congress

**An Uncertainty-Based Corner Case Detection Framework for Autonomous Driving**

2021

Sensors &amp; IIoT - Manufacturing, Automation and Robotics

**Model-Enhanced Machine Learning for Astronomical Discovery**

2016

University of Toronto

**Tests of the Planetary Hypothesis for PTFO 8-8695b**

2015

Boston University

**Tests of the Planetary Hypothesis for PTFO 8-8695b**

2015

Harvard University

**SELECTED PUBLICATIONS**

**Yu, L.;** Vanderburg, A.; Huang, C. X.; Shallue, C. J.; Crossfield, I. J. M. et al. (2019), *Identifying Exoplanets with Deep Learning III: Automated Triage and Vetting of TESS Candidates*, the *Astronomical Journal*, 158, 25.

**Yu, L.;** Zhou, G.; Rodriguez, J. E.; Huang, C. X.; Vanderburg, A. et al. (2018), *EPIC 246851721 b: A Tropical Jupiter Transiting a Rapidly Rotating Star in a Well-Aligned Orbit*, the *Astronomical Journal*, 156, 250.

**Yu, L.;** Rodriguez, J. E.; Eastman, J. D.; Crossfield, I. J. M.; Shporer, A. et al. (2018), *Two Warm, Low-density Sub-Jovian Planets Orbiting Bright Stars in K2 Campaigns 13 and 14*, the *Astronomical Journal*, 156, 127.

**Yu, L.;** Crossfield, I. J. M.; Schlieder, J. E.; Kosiarek, M. R.; Feinstein, A. D. et al. (2018), *Planetary Candidates from K2 Campaign 16*, the *Astronomical Journal*, 156, 22.

**Yu, L.;** Winn, J. N.; Gillon, M.; Albrecht, S.; Rappaport, S. et al. (2015), *Tests of the Planetary Hypothesis for PTFO 8-8695b*, the *Astrophysical Journal*, 812, 48.

**Yu L.;** Nelson, K.; Nagai, D. (2015), *The Influence of Mergers on Scatter and Evolution in Sunyaev-Zel'dovich Effect Scaling Relations*, the *Astrophysical Journal*, 807, 12.